Given: Due:	Further Maths (Core Pure) HW 11	Name: MARK SCHEME

Chapter 7: Matrix transformation

1)

$$\mathbf{A} = \begin{pmatrix} 2 & -1 \\ 4 & 3 \end{pmatrix}, \quad \mathbf{P} = \begin{pmatrix} 3 & 6 \\ 11 & -8 \end{pmatrix}$$

(a) Find $\mathbf{A}^{-1}$

(2)

The transformation represented by the matrix **B** followed by the transformation represented by the matrix **A** is equivalent to the transformation represented by the matrix **P**.

(b) Find **B**, giving your answer in its simplest form.

(3)

In part (a) it was rare to see an error. However part (b) proved to challenge a substantial minority, who multiplied their matrices in the wrong order. Most candidates spotted that they could use their inverse matrix to quickly find matrix ***B***, but some did use the much longer method; pre-multiplying by ***A***, then solving simultaneous equations.

Question Number	Scheme		Marks
(a)	$\mathbf{A}=\begin{pmatrix} 2 & -1 \\ 4 & 3 \end{pmatrix}, \mathbf{P}=\begin{pmatrix} 3 & 6 \\ 11 & -8 \end{pmatrix}$		
	$\mathbf{A}^{-1}=\frac{1}{10}\begin{pmatrix} 3 & 1 \\ -4 & 2 \end{pmatrix}$	Either $\frac{1}{10}$ or $\begin{pmatrix} 3 & 1 \\ -4 & 2 \end{pmatrix}$	M1
		Correct matrix seen.	A1
			[2]
(b) Way 1	$\mathbf{P}=\mathbf{AB}$ $\Rightarrow \mathbf{A}^{-1}\mathbf{P}=\mathbf{A}^{-1}\mathbf{AB} \Rightarrow \mathbf{B}=\mathbf{A}^{-1}\mathbf{P}$		
	$\mathbf{B} = \frac{1}{10}\begin{pmatrix} 3 & 1 \\ -4 & 2 \end{pmatrix}\begin{pmatrix} 3 & 6 \\ 11 & -8 \end{pmatrix}$	Multiplies **their** $\mathbf{A}^{-1}$ by **P** in correct order. This substituted statement is sufficient.	M1
	$=\begin{pmatrix} 2 & 1 \\ 1 & -4 \end{pmatrix}$	At least 2 elements correct or $k\begin{pmatrix} 20 & 10 \\ 10 & -40 \end{pmatrix}$ oe. May be unsimplified	A1
		Correct simplified matrix.	A1
			[3]
(b) Way 2	$\{\mathbf{P}=\mathbf{AB}\Rightarrow\}$		
	$\begin{pmatrix} 3 & 6 \\ 11 & -8 \end{pmatrix}=\begin{pmatrix} 2 & -1 \\ 4 & 3 \end{pmatrix}\begin{pmatrix} a & b \\ c & d \end{pmatrix}$		
	$\begin{pmatrix} 3 & 6 \\ 11 & -8 \end{pmatrix}=\begin{pmatrix} 2a-c & 2b-d \\ 4a+3c & 4b+3d \end{pmatrix}$	Attempt to multiply **A** by **B** in the correct order and puts equal to **P**	M1
	$\Rightarrow a=2, c=1, b=1, d=-4$		
	So, $\mathbf{B}=\begin{pmatrix} 2 & 1 \\ 1 & -4 \end{pmatrix}$	At least 2 elements are correct.	A1
		Correct matrix.	A1
			[3] 5

2) (i) In each of the following cases, find a 2 × 2 matrix that represents

(a) a reflection in the line $y = -x$,
(b) a rotation of 135° anticlockwise about (0, 0),
(c) a reflection in the line $y = -x$ followed by a rotation of 135° anticlockwise about (0, 0).

(4)

(ii) The triangle T has vertices at the points $(1, k)$, $(3, 0)$ and $(11, 0)$, where k is a constant.

Triangle T is transformed onto the triangle T' by the matrix

$$\begin{pmatrix} 6 & -2 \\ 1 & 2 \end{pmatrix}$$

Given that the area of triangle T' is 364 square units, find the value of k.

(6)

Many students in question (i) could not find the matrices securely to represent the basic transformations in question (a) and question (b). The correct order of composite transformation in question (c) was found to be challenging with many students attempting to multiply in the wrong order. Some students in question (b) and question (c) left the matrix in terms of the trigonometrical ratios. In question (ii) the majority of students successfully calculated the value of the determinant. Of those who went on to calculate an area for the triangle many used 364 and their determinant and were successful in finding a value for *k*. Those who attempted to find the area of the transformed triangle in terms of *k* usually made no further progress.

Question Number	Scheme	Notes	Marks
(i)(a)	$\begin{pmatrix} 0 & -1 \\ -1 & 0 \end{pmatrix}$		B1
(b)	$\begin{pmatrix} -\frac{1}{\sqrt{2}} & -\frac{1}{\sqrt{2}} \\ \frac{1}{\sqrt{2}} & -\frac{1}{\sqrt{2}} \end{pmatrix}$		B1
(c)	$\begin{pmatrix} -\frac{1}{\sqrt{2}} & -\frac{1}{\sqrt{2}} \\ \frac{1}{\sqrt{2}} & -\frac{1}{\sqrt{2}} \end{pmatrix}\begin{pmatrix} 0 & -1 \\ -1 & 0 \end{pmatrix}=\begin{pmatrix} \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} \\ \frac{1}{\sqrt{2}} & -\frac{1}{\sqrt{2}} \end{pmatrix}$	M1: Multiplies their (b) x their (a) in the correct order A1: Correct matrix Correct matrix seen M1A1	M1A1
			(4)
(ii)	Area triangle T $= \frac{1}{2}\times(11-3)\times k = 4k$	M1: Correct method for area for *T* A1: 4k	M1A1
	$\det\begin{pmatrix} 6 & -2 \\ 1 & 2 \end{pmatrix} = 6\times2-1\times(-2)(=14)$	M1: Correct method for determinant A1: 14	M1A1
	Area triangle $T = \frac{364}{"14"}(=26) \Rightarrow 4k = 26$	Uses 364 and their determinant correctly to form an equation in *k*.	M1
	$k = \frac{26}{4}\left(=\frac{13}{2}\right)$	Accept $k = \pm\frac{13}{2}$ or $k = -\frac{13}{2}$	A1
			(6)
			Total 10

3)

$$\mathbf{A} = \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}, \quad \mathbf{B} = \begin{pmatrix} 2 & 3 \\ 1 & 4 \end{pmatrix}$$

The transformation represented by **B** followed by the transformation represented by **A** is equivalent to the transformation represented by **P**.

(a) Find the matrix **P**.

(2)

Triangle *T* is transformed to the triangle *T* ' by the transformation represented by **P**.

Given that the area of triangle *T* ' is 24 square units,

(b) find the area of triangle *T*.

(3)

Triangle *T* ' is transformed to the original triangle *T* by the matrix represented by **Q**.

(c) Find the matrix **Q**.

(2)

A most common error in part (a) was to find **P** as **BA** instead of **AB**, but candidates were awarded follow through marks in the later parts of the question. The value of the determinant was usually correct, and most knew that this helped to find the area, but some candidates multiplied by their determinant in part (b) rather than dividing. In part (c) a few candidates failed to realise that $\mathbf{P}^{-1}$ was being asked for, but many were able to find the inverse correctly from their answer to part (a).

Question Number	Scheme		Marks
	$\mathbf{A} = \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}$, $\mathbf{B} = \begin{pmatrix} 2 & 3 \\ 1 & 4 \end{pmatrix}$		
(a)	$\mathbf{P} = \mathbf{AB}\left\{= \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}\begin{pmatrix} 2 & 3 \\ 1 & 4 \end{pmatrix}\right\}$	$\mathbf{P} = \mathbf{AB}$, seen or implied.	M1
	$\mathbf{P} = \begin{pmatrix} 1 & 4 \\ -2 & -3 \end{pmatrix}$	Correct answer.	A1
			[2]
(b)	$\det \mathbf{P} = 1(-3) - (4)(-2)\ \{= -3 + 8 = 5\}$	Applies "$ad - bc$".	M1
	$\text{Area}(T) = \underline{\frac{24}{5}}\ (\text{units})^2$	$\frac{24}{\text{their } \det \mathbf{P}}$, dependent on previous M	dM1
		$\underline{\frac{24}{5}}$ or $\underline{4.8}$	A1ft
			[3]
(c)	$\mathbf{QP} = \mathbf{I} \Rightarrow \mathbf{QPP}^{-1} = \mathbf{IP}^{-1} \Rightarrow \mathbf{Q} = \mathbf{P}^{-1}$		
	$\mathbf{Q} = \mathbf{P}^{-1} = \frac{1}{5}\begin{pmatrix} -3 & -4 \\ 2 & 1 \end{pmatrix}$	$\mathbf{Q} = \mathbf{P}^{-1}$ stated or an attempt to find $\mathbf{P}^{-1}$.	M1
		Correct ft inverse matrix.	A1ft
			[2] 7
	Using **BA**, area is the same in (b) and inverse is $\frac{1}{5}\begin{pmatrix} 1 & -2 \\ 4 & -3 \end{pmatrix}$ in (c) and could gain ft marks.		

4) The transformation U, represented by the 2 × 2 matrix **P**, is a rotation through 90º anticlockwise about the origin.

(a) Write down the matrix **P**.

(1)

The transformation V, represented by the 2 × 2 matrix **Q**, is a reflection in the line $y = -x$.

(b) Write down the matrix **Q**.

(1)

Given that U followed by V is transformation T, which is represented by the matrix **R**,

(c) express **R** in terms of **P** and **Q**,

(1)

(d) find the matrix **R**,

(2)

(e) give a full geometrical description of T as a single transformation.

(2)

This question usually contained some errors from candidates. In (a) the matrix was usually correct, but in (b) the matrix attracted more errors. In (c) and (d) many candidates did not multiply their matrices in the correct order, however a large number of candidates made a reasonable attempt at the matrix multiplication and gained at least the method mark. (e) could be answered without reference to the matrix obtained in (d) and for many this was a useful approach. A single transformation was required in (e) and most got at least one mark here. A common error was to choose the wrong axis.

Question Number	Scheme	Marks
	(a) $\begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}$	B1 **(1)**
	(b) $\begin{pmatrix} 0 & -1 \\ -1 & 0 \end{pmatrix}$	B1 **(1)**
	(c) $\mathbf{R} = \mathbf{QP}$	B1 **(1)**
	(d) $\mathrm{R} = \begin{pmatrix} 0 & -1 \\ -1 & 0 \end{pmatrix}\begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix} = \begin{pmatrix} -1 & 0 \\ 0 & 1 \end{pmatrix}$	M1 A1 cao **(2)**
	(e) Reflection in the y axis	B1 B1 **(2)** [7]
Notes	(a) and (b) Signs must be clear for B marks. (c) Accept **QP** or their 2x2 matrices in the correct order only for B1. (d) M for their **QP** where answer involves ±1 and 0 in a 2x2 matrix. A for correct answer only. (e) First B for Reflection, Second B for 'y axis' or 'x=0'. Must be single transformation. Ignore any superfluous information.	

5)

$$\mathbf{A} = \begin{pmatrix} -\frac{1}{2} & -\frac{\sqrt{3}}{2} \\ \frac{\sqrt{3}}{2} & -\frac{1}{2} \end{pmatrix}$$

(a) Describe fully the single geometrical transformation U represented by the matrix **A**.

(3)

The transformation V, represented by the 2×2 matrix **B**, is a reflection in the line $y = -x$

(b) Write down the matrix **B**.

(1)

Given that U followed by V is the transformation T, which is represented by the matrix **C**,

(c) find the matrix **C**.

(2)

(d) Show that there is a real number k for which the point $(1, k)$ is invariant under T.

(4)

Parts (a), (b), (c) should have been routine procedures and it was surprising how any students made mistakes in these parts.

In part (a), the majority of students realised that the transformation was a rotation but struggled with the angle, whilst many students failed to mention that the rotation was about the origin. A significant number also lost all three marks by attempting to describe a composite transformation.

Part (b) provided mixed responses. Many did state the correct matrix, but a variety of incorrect matrices were also seen - the entries on the wrong diagonals being the most common. The use of a sketch would be advised to help trace where the coordinates (1, 0) and (0, 1) map to under the transformation, and hence help establish the required matrix.

Many students did manage to get full marks in (c) due to the follow through mark, however there was a surprising number who mixed up the order of matrix multiplication and hence gained no marks.

Part (d) was, on the whole, well attempted with only the final mark causing some difficulty. Most students that attempted (d) managed to achieve M1A1ft, although errors in (b) or (c) prevented them getting any further. Students that had gained full marks in (b) and (c) often went on to get the next A mark for finding a correct value for k but not many went on to use the second equation to confirm this, thus losing the final B mark.

Question	Scheme	Marks	AOs
(a)	Rotation	B1	1.1b
	120 degrees (anticlockwise) or $\frac{2\pi}{3}$ radians (anticlockwise) Or 240 degrees clockwise or $\frac{4\pi}{3}$ radians clockwise	B1	2.5
	About (from) the origin. Allow (0, 0) or O for origin.	B1	1.2
		(3)	
(b)	$\begin{pmatrix} 0 & -1 \\ -1 & 0 \end{pmatrix}$	B1	1.1b
		(1)	
(c)	$\begin{pmatrix} 0 & -1 \\ -1 & 0 \end{pmatrix}\begin{pmatrix} -\frac{1}{2} & -\frac{\sqrt{3}}{2} \\ \frac{\sqrt{3}}{2} & -\frac{1}{2} \end{pmatrix} = \begin{pmatrix} \dots & \dots \\ \dots & \dots \end{pmatrix}$	M1	1.1b
	$\begin{pmatrix} 0 & -1 \\ -1 & 0 \end{pmatrix}\begin{pmatrix} -\frac{1}{2} & -\frac{\sqrt{3}}{2} \\ \frac{\sqrt{3}}{2} & -\frac{1}{2} \end{pmatrix} = \begin{pmatrix} -\frac{\sqrt{3}}{2} & \frac{1}{2} \\ \frac{1}{2} & \frac{\sqrt{3}}{2} \end{pmatrix}$	A1ft	1.1b
		(2)	
(d)	$\begin{pmatrix} -\frac{\sqrt{3}}{2} & \frac{1}{2} \\ \frac{1}{2} & \frac{\sqrt{3}}{2} \end{pmatrix}\begin{pmatrix} 1 \\ k \end{pmatrix} = \begin{pmatrix} 1 \\ k \end{pmatrix} = \dots$ or $\begin{pmatrix} -\frac{\sqrt{3}}{2} & \frac{1}{2} \\ \frac{1}{2} & \frac{\sqrt{3}}{2} \end{pmatrix}\begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} x \\ y \end{pmatrix} = \dots$ Note: $\begin{pmatrix} -\frac{\sqrt{3}}{2} & \frac{1}{2} \\ \frac{1}{2} & \frac{\sqrt{3}}{2} \end{pmatrix}\begin{pmatrix} 1 \\ k \end{pmatrix} = \begin{pmatrix} -\frac{\sqrt{3}}{2}+\frac{1}{2}k \\ \frac{1}{2}+\frac{\sqrt{3}}{2}k \end{pmatrix} = \begin{pmatrix} 1 \\ k \end{pmatrix}$ can score M1 (for the matrix equation) but needs an equation to be "extracted" to score the next A1	M1	3.1a
	$-\frac{\sqrt{3}}{2}+\frac{1}{2}k=1$ or $\frac{1}{2}+\frac{\sqrt{3}}{2}k=k$ or $x=-\frac{\sqrt{3}}{2}x+\frac{1}{2}y$ or $y=\frac{1}{2}x+\frac{\sqrt{3}}{2}y$ (Note that candidates may then substitute x = 1 which is acceptable)	A1ft	1.1b
	$-\frac{\sqrt{3}}{2}+\frac{1}{2}k=1$ or $x=-\frac{\sqrt{3}}{2}x+\frac{1}{2}y \Rightarrow k=2+\sqrt{3}\left(\text{or } \frac{1}{2-\sqrt{3}}\right)$	A1	1.1b
	$\frac{1}{2}+\frac{\sqrt{3}}{2}k=k$ or $y=\frac{1}{2}x+\frac{\sqrt{3}}{2}y \Rightarrow k=2+\sqrt{3}\left(\text{or } \frac{1}{2-\sqrt{3}}\right)$	B1	1.1b
		(4)	
			(10 marks)

Notes
(a) B1: Identifies the transformation as a rotation B1: Correct angle. Allow equivalents in degrees or radians. B1: Identifies the origin as the centre of rotation These marks can only be awarded as the elements of a **single transformation** (b) B1: Shows the correct matrix in the correct form (c) M1: Multiplies the matrices in the correct order (evidence of multiplication can be taken from 3 correct or 3 correct ft elements) A1ft: Correct matrix (follow through their matrix from part (b)) A correct matrix or a correct follow through matrix implies both marks. (d) M1: Translates the problem into a matrix multiplication to obtain at least one equation in k or in x and y A1ft: Obtains one correct equation (follow through their matrix from part (c)) A1: Correct value for k in any form B1: Checks their answer by independently solving both equations correctly to obtain $2+\sqrt{3}$ both times or substitutes $2+\sqrt{3}$ into the other equation to confirm its validity

6)

$$\mathbf{P} = \begin{pmatrix} -\frac{1}{\sqrt{2}} & -\frac{1}{\sqrt{2}} \\ \frac{1}{\sqrt{2}} & -\frac{1}{\sqrt{2}} \end{pmatrix}$$

(a) Describe fully the single geometrical transformation U represented by the matrix **P**.

(2)

The transformation U maps the point A, with coordinates (p, q), onto the point B, with coordinates $(6\sqrt{2}, 3\sqrt{2})$.

(b) Find the value of p and the value of q.

(3)

The transformation V, represented by the 2×2 matrix **Q**, is a reflection in the line with equation $y = x$.

(c) Write down the matrix **Q**.

(1)

The transformation U followed by the transformation V is the transformation T. The transformation T is represented by the matrix **R**.

(d) Find the matrix **R**.

(3)

(e) Deduce that the transformation T is self-inverse.

(1)

Many correct solutions were seen to part (a) and nearly all students recognised this as a rotation matrix. The most common error was to be omission of the centre as part of their description.

A few students used an angle of 45° instead of 135° and even fewer used the wrong direction of rotation.

Most students obtained the correct solution in part (b). The most popular approach was to use $\mathbf{P}\begin{pmatrix} p \\ q \end{pmatrix} = \begin{pmatrix} 6\sqrt{2} \\ 3\sqrt{2} \end{pmatrix}$, multiply out and solve the resulting simultaneous equations. Any errors were caused by incorrect manipulation of the surds. A minority did take the more straightforward route of finding the inverse of matrix **P** and pre-multiplying the position vector $\begin{pmatrix} 6\sqrt{2} \\ 3\sqrt{2} \end{pmatrix}$ by it to directly produce values for p and q.

A majority of correct matrices were seen in part (c). A tiny number used identity matrices or else –1 instead of 1.

Mostly correct products found in part (d), although a minority did find PQ instead of QP.

Many students did not fully understand "self-inverse" in part (e) and simply found the determinant as a sufficient argument. The most popular successful approach was to find $\mathbf{R}^{-1}$ and show it was the same as **R**. Some students successfully showed that $\mathbf{R}^2$ was the identity matrix.

Question Number	Scheme	Marks
	(a) **Rotation** ,135 degrees or $\frac{3\pi}{4}$ radians (anticlockwise) about O or 225 degrees or $\frac{5\pi}{4}$ clockwise about O.	M1, A1 (2)
	(b) $\begin{pmatrix} -\frac{1}{\sqrt{2}} & -\frac{1}{\sqrt{2}} \\ \frac{1}{\sqrt{2}} & -\frac{1}{\sqrt{2}} \end{pmatrix}\begin{pmatrix} p \\ q \end{pmatrix} = \begin{pmatrix} 6\sqrt{2} \\ 3\sqrt{2} \end{pmatrix}$	
	$-p-q=12$ and $p-q=6$ or equivalent	M1A1
	$p=-3$ and $q=-9$ or $\begin{pmatrix} -3 \\ -9 \end{pmatrix}$	B1 cso (3)
	ALT Uses Inverse matrix $\mathbf{P}^{-1}$ with vector $= \begin{pmatrix} -\frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} \\ -\frac{1}{\sqrt{2}} & -\frac{1}{\sqrt{2}} \end{pmatrix}\begin{pmatrix} 6\sqrt{2} \\ 3\sqrt{2} \end{pmatrix}$	M1A1
	$p=-3$ and $q=-9$ or $\begin{pmatrix} -3 \\ -9 \end{pmatrix}$	B1 cso (3)
	(c) $\mathbf{Q} = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$	B1 (1)
	Accept **T** if used instead of **R**	
	(d) $\mathbf{R} = "\mathbf{Q}"\mathbf{P} = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}\begin{pmatrix} -\frac{1}{\sqrt{2}} & -\frac{1}{\sqrt{2}} \\ \frac{1}{\sqrt{2}} & -\frac{1}{\sqrt{2}} \end{pmatrix} = \begin{pmatrix} \frac{1}{\sqrt{2}} & -\frac{1}{\sqrt{2}} \\ -\frac{1}{\sqrt{2}} & -\frac{1}{\sqrt{2}} \end{pmatrix}$	M1 A1 A1 (3)
	(e) $\mathbf{R}^{-1} = \frac{1}{-1}\begin{pmatrix} -\frac{1}{\sqrt{2}} & +\frac{1}{\sqrt{2}} \\ +\frac{1}{\sqrt{2}} & +\frac{1}{\sqrt{2}} \end{pmatrix} = \begin{pmatrix} \frac{1}{\sqrt{2}} & -\frac{1}{\sqrt{2}} \\ -\frac{1}{\sqrt{2}} & -\frac{1}{\sqrt{2}} \end{pmatrix}$	
	$= \mathbf{R}$ (so matrix is self inverse and so transformation is self inverse)	B1 **(1)** **10 marks**
	ALT 1 (e) $\mathbf{RR} = \begin{pmatrix} \frac{1}{\sqrt{2}} & -\frac{1}{\sqrt{2}} \\ -\frac{1}{\sqrt{2}} & -\frac{1}{\sqrt{2}} \end{pmatrix}\begin{pmatrix} \frac{1}{\sqrt{2}} & -\frac{1}{\sqrt{2}} \\ -\frac{1}{\sqrt{2}} & -\frac{1}{\sqrt{2}} \end{pmatrix} = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$ (so **R** is self inverse and so transformation is self inverse)	B1
	ALT 2 (e) Matrix represents a reflection (so is self inverse)	B1

Notes

(a) M1: Rotation only A1: 135 degrees about O
SC: 135 degrees about O only award M1A0.
(b) M1: Multiplies matrices in correct order to obtain two equations in p and q.
A1: Two correct equations
B1 cso: p and q both correct, may be in vector form. No errors seen in solution.
ALT (b) M1: Attempt to find Inverse Matrix and pre-multiply A1: Correct Inverse Matrix used
B1 cso: p and q both correct, may be in vector form. No errors seen in solution.
(d) M1: Sets matrix product correct way round and obtains one correct term for their **Q**
A1: Two correct terms from a correct **Q**. **Q** incorrect award A0 here. A1: Completely correct matrix
(e) B1: Calculates $\mathbf{R}^{-1}$ and indicates that $\mathbf{R}^{-1} = \mathbf{R}$ or calculates $\mathbf{R}^2$ and indicates that $\mathbf{R}^2 = \mathbf{I}$ or states that **R** represents a reflection.